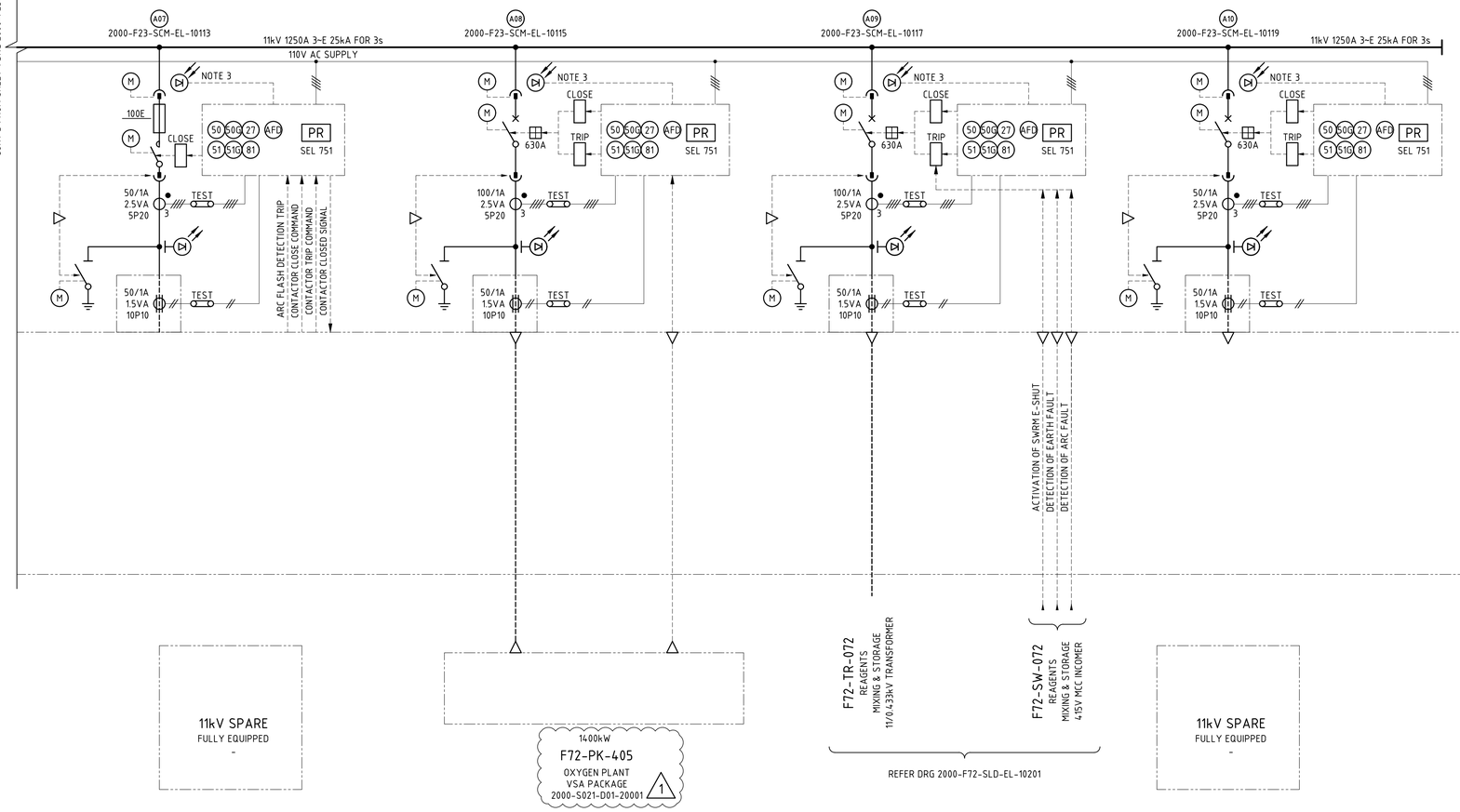
F23-SR-001
FINAL TAILS PUMPS 11kV & 415V SWITCHROOM

F23-SW-016
FINAL TAILS PUMPS 11kV SWITCHBOARD



	Approved	[X]
	Approved as Noted	[ ]
	Not Approved	[ ]
	For Information Only	[ ]

Signed [redacted] Date 15/10/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

NOTES

1. THE SWITCHBOARD SHALL BE EXTENSIBLE ONE END FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER AS SHOWN.
2. ARC FLASH IN INCOMER CIRCUIT BREAKER COMPARTMENT IS NOT CURRENT SUPERVISED.
   TRIPPING OF BUS AND DOWNSTREAM CB/FC COMPARTMENTS IS CURRENT SUPERVISED.
3. ARC FLASH POINT DETECTORS SHALL BE PLACED IN THE CABLE ZONE, CIRCUIT BREAKER CHAMBER OR FUSE CONTACTOR CHAMBER AND THE BUSBAR CHAMBER. THE DETECTORS SHALL BE CONNECTED DIRECTLY TO THE PROTECTION RELAY IN EACH PANEL.
4. ONE OFF 30/0.05A PROTECTION CT SHALL BE INSTALLED IN THE NER CUBICLE AND CONNECTED TO THE INCOMER PROTECTION RELAY TO MONITOR FOR EARTH FAULT DETECTION.
5. SWITCHROOM EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON IS HARDWIRED DIRECT TO THE SHUNT TRIP OF THE INCOMER CIRCUIT BREAKER AND UPSTREAM 33kV FEEDER.
6. 33/11kV TRANSFORMER INSTRUMENTATION TRIP CONTACTS SHALL BE HARDWIRED TO THE 11kV SWITCHBOARD INCOMER PROTECTION RELAY WHICH SHALL INTERTRIP THE UPSTREAM 33kV FEEDER.
7. UPSTREAM 33kV FEEDER SHALL BE TRIPPED ON THE EARTH & ARC FAULT DETECTION OF THE DOWNSTREAM INCOMER ACB.

LEGEND

PR	INCOMER / FEEDER PROTECTION RELAY
PQM	POWER QUALITY METER
	ARC FLASH SENSOR (POINT TYPE) NOTES 2 & 3
	CAPACITIVE VOLTAGE INDICATION
	MECHANICAL INTERLOCK DEVICE
E-SHUT	SWITCHROOM EMERGENCY SHUTDOWN
M	MOTOR OPERATOR
27	UNDER VOLTAGE
50	INSTANTANEOUS OVER-CURRENT
50G 50N	INSTANTANEOUS EARTH/NEUTRAL OVERCURRENT
51	TIMED OVER-CURRENT
51G 51N	TIMED EARTH/NEUTRAL OVERCURRENT
59	OVER VOLTAGE
64	EARTH FAULT
49	TRANSFORMER TEMPERATURE
63	TRANSFORMER PRESSURE / BUCCHOLZ
71	TRANSFORMER LIQUID / GAS LEVEL
81	FREQUENCY
AFD	ARC FLASH DETECTION

DRG No.	REFERENCES
2000-F23-DRG-EL-10001	F23-SR-001 11kV FINAL TAILS PUMPS SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT
2000-F23-SLD-EL-10051	F23-SR-001 11kV FINAL TAILS PUMPS SWITCHBOARD KEY INTERLOCKING SINGLE LINE DIAGRAM
2000-F23-SLD-EL-10001	F23-SR-001 11kV FINAL TAILS PUMPS SWITCHBOARD SINGLE LINE DIAGRAM SHEET 1 OF 2
2000-F00-SLD-EL-10001	OVERALL SITEWIDE SINGLE LINE DIAGRAM
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
1	19SEP25	FAT REDLINES INCORPORATED					
0	22OCT24	APPROVED FOR CONSTRUCTION					

NORTHERN STAR RESOURCES LIMITED

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PRIMERO PROJECT NUMBER: 23517

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: FINAL TAILINGS HANDLING & STORAGE
F23-SW-016 11kV SWITCHBOARD
SINGLE LINE DIAGRAM SHEET 2

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F23-SLD-EL-10002	1

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F23 - FINAL TAILINGS HANDLING & STORAGE\2000-F23-SLD-EL-10002.DWG 02OCT25 9:04AM HOTSEAT04